~~HARTLEY AND~~ COLPITT'S OSCILLATOR.

AIM:

To design and construct the ~~Hartley and~~ Colpitt's oscillator and verify its theoretical and practical frequencies.

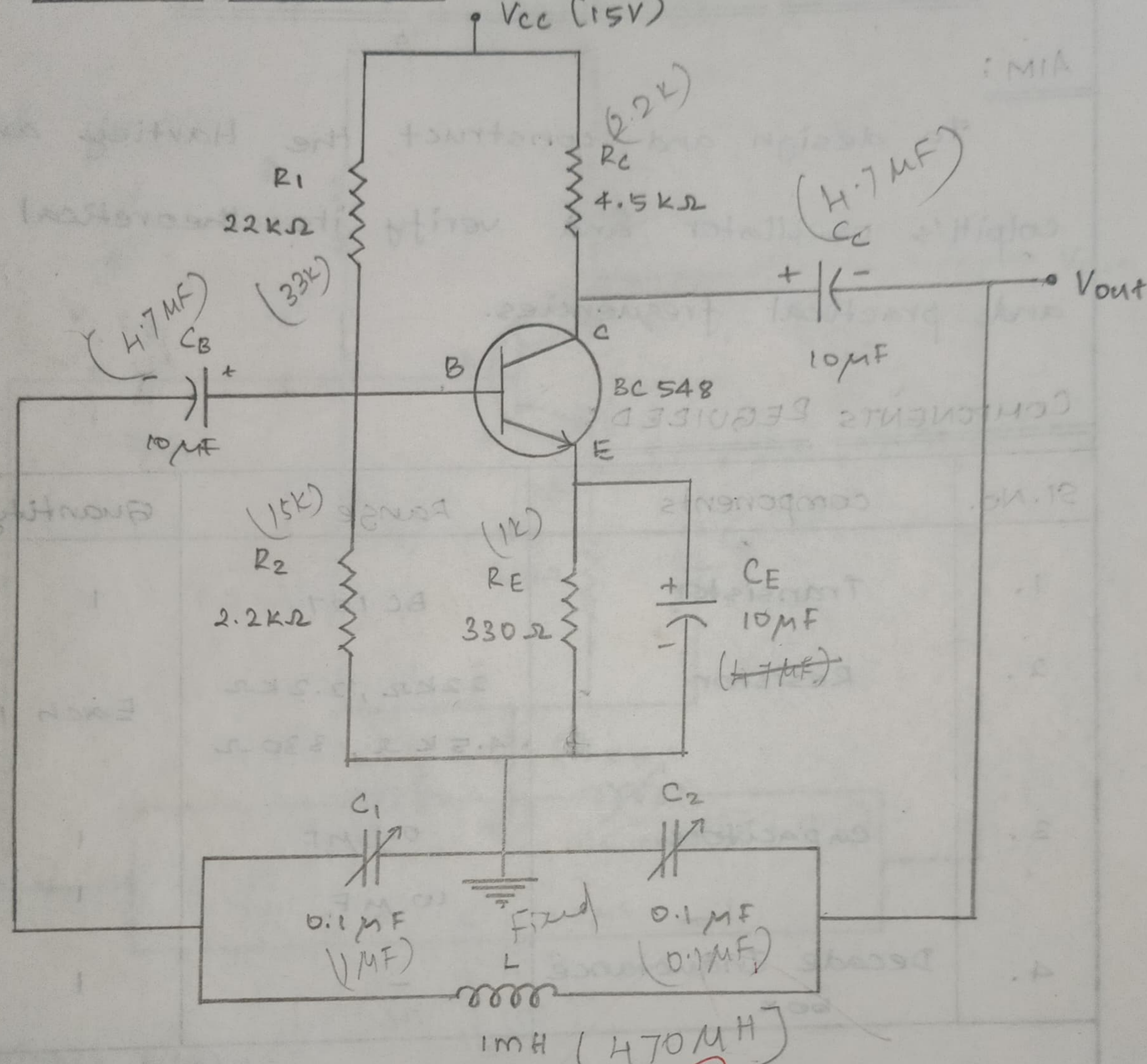
COMPONENTS REQUIRED:

Sl. No.	components	Range	Quantity
1.	Transistor	BC 107	1
2.	Resistor	22kΩ, 2.2kΩ 4.5kΩ, 330Ω	Each 1
3.	capacitors	0.1 μF 10 μF	1 1
4.	~~Decade Inductance box~~	–	1

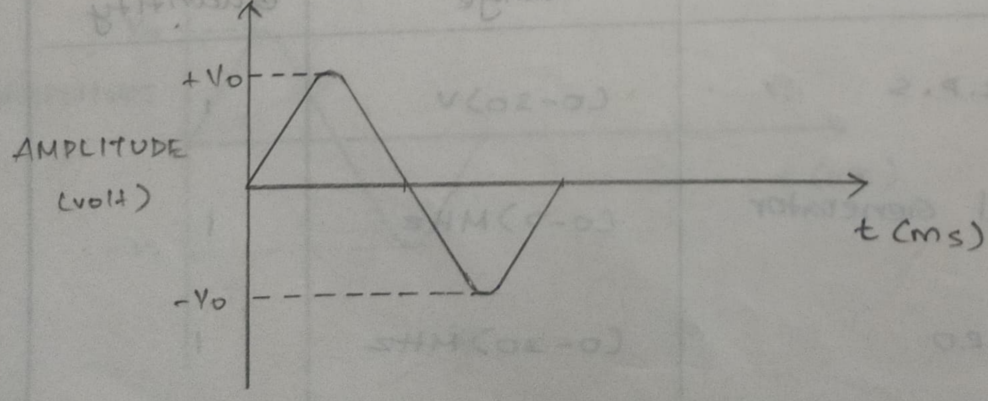
APPARATUS REQUIRED:

Sl. No.	apparatus	Range	Quantity
1.	R.P.S	(0-30)V	1
2.	signal Generator	(0-2)MHz	1
3.	CRO	(0-30)MHz	1

COLPITT'S OSCILLATOR:



MODEL GRAPH:



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NAME :

Expt No :
Page No :

FORMULA :

i) $I_B = \frac{I_C}{\beta}$.

ii) $R_C = \frac{V_{CC} - V_{CE}}{I_C} - R_E$

$V_{CC} - V_{CE} = I_C R_C + I_E R_E$

$V_{CE} = V_{CC} - I_C (R_C + R_E)$

iii) $I_E = I_B + I_C$.

iv) $V_{TH} = I_B R_B + V_{BE} + I_E R_E$.

v) $R_{TH} = \frac{(1-S)(1+\beta)}{(1-S-\beta)} \cdot R_E$.

vi) $R_1 = \frac{V_{CC}}{V_{TH}} \cdot R_{TH}$.

vii) $R_2 = \frac{V_{TH}}{V_{CC} - V_{TH}} \cdot R_1$

~~HARTLEY OSCILLATOR :~~

$\cancel{L_{eq} = L_1 + L_2}$

$\cancel{f = \frac{1}{2\pi \sqrt{L_{eq} \cdot C}}}$

COLPITT'S OSCILLATOR :

$C_{eq} = \frac{C_1 \cdot C_2}{C_1 + C_2}$

$f = \frac{1}{2\pi \sqrt{L \cdot C_{eq}}}$.

COLPITT'S OSCILLATOR:

S. No.	WAVEFORM	AMPLITUDE (V)	TIME (ms)	FREQUENCY (Hz).
1.	Output signal	2.6 x 10 V	24.34 KHZ 1x20 MS	25 KHZ 50 kHz.

24kHZ

RGCET DATE:	NAME :	Expt No : Page No :

THEORY

COLPITT'S OSCILLATOR:

Colpitt's oscillator is same as that of the Hartley oscillator except that the emitter is connected between the capacitance C_1 and C_2. The resistors R_1, R_2 and R_{ie} provides the necessary biasing conditions. The functions of C_c is to block d.c. current and to provide an a.c. path. The radio frequency choke offers very high impedence to high frequency current. Thus it prevents radio frequency current from reaching the collector.

3Y

The ~~frequency~~ determing network is a parallel combination resonant circuit consisting of capacitor C_1 and C_2 and inductor L. The voltage developed across C_1 provides the regenerative feedback required for the sustained oscillations. when the collector supply voltage is switched ON, transient current is produced in the tank circuit.

COLPITT'S OSCILLATOR:

$L = 200\,\mu H \qquad C_1 = 0.1\,\mu F \qquad C_2 = 0.1\,\mu F$

$$C_{eq} = \frac{0.1 \times 0.1 \times 10^{-12}}{0.2 \times 10^{-6}} = 5 \times 10^{-8}$$

$$f = \frac{1}{2\pi\sqrt{L.C_{eq}}}$$

$$= \frac{1}{2\pi\sqrt{200 \times 10^{-6} \times 5 \times 10^{-8}}}$$

$$f = 50.33\,KHz$$

PROCEDURE :

1. Connections are given as per the circuit diagrams.

2. when an supply voltage V_{cc} is applied to the circuit, and note the corresponding output voltage.

3. the same procedure is repeated for the colpitt's oscillator.

4. The output waveform is obtained and noted.

5. A graph is drawn for the output waveforms of both Hartley and colpitt's oscillator.

RESULT :

23/10/07

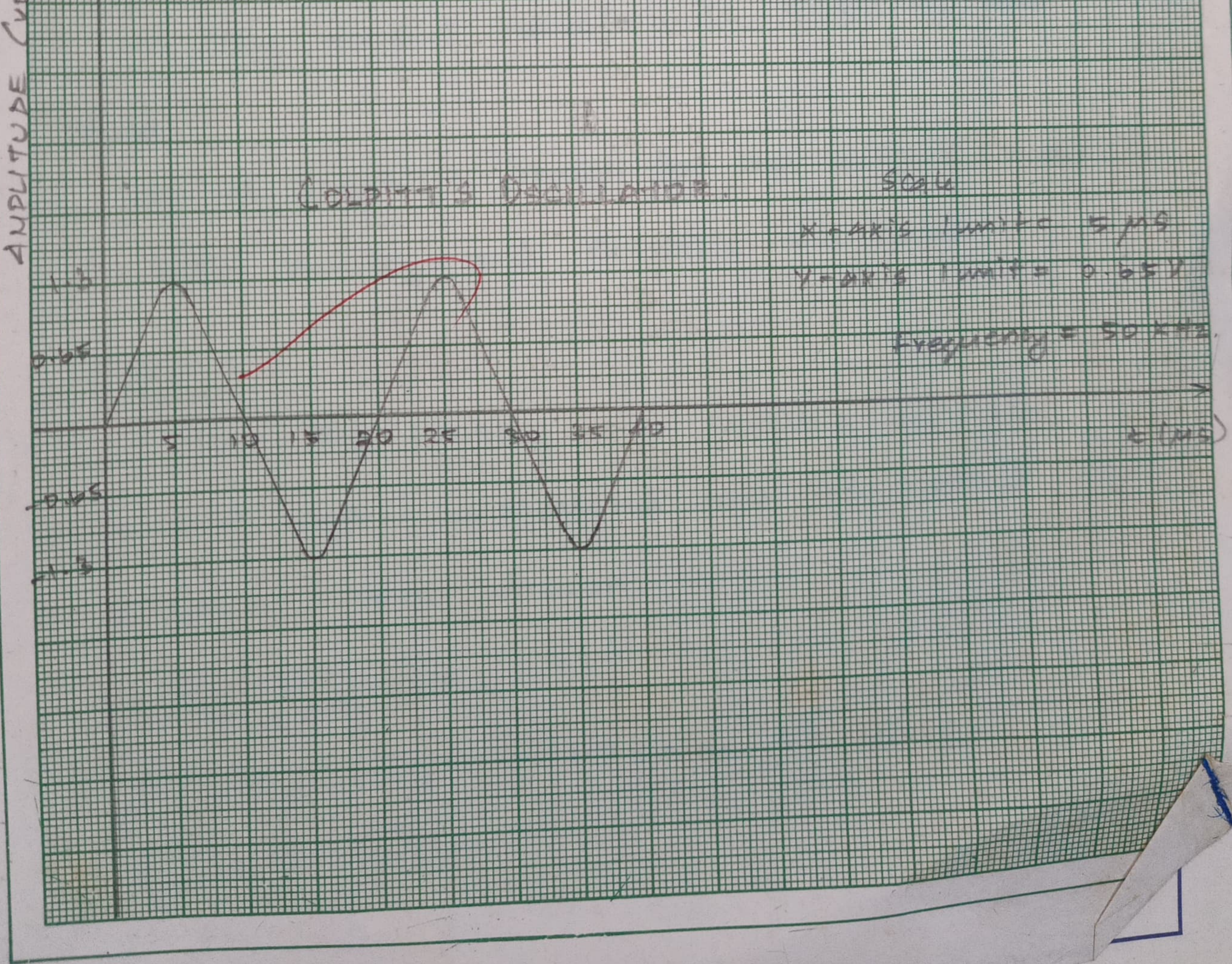
Thus ~~the~~ Hartley and colpitt's oscillators were designed and constructed and its output waveform were drawn.

Hartley Oscillator :

Theoretical Frequency = 1.59 kHz. Practical Freq = 1.33 kHz

Colpitt's Oscillator :

Theoretical Frequency = 50.33 kHz. Practical Freq = 50 kHz


AMPLITUDE (V)
COLPITT'S OSCILLATOR
Scale
X-Axis 1 unit = 5 μs
Y-Axis 1 unit = 0.65 V
Frequency = 50 KHz
1.3
0.65
-0.65
-1.3
5
10
15
20
25
30
35
40
t (μs)